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# Step 1: Import necessary libraries
from sklearn import tree
import matplotlib.pyplot as plt

# Step 2: Prepare the data
# Features:
# Study Hours
# Sleep Hours
# Tuition (0=No, 1=Yes)

# Each row = [Study Hours, Sleep Hours, Tuition]
X = [
    [2, 6, 0],
    [3, 7, 1],
    [1, 5, 0],
    [4, 8, 1],
    [2, 7, 1],
    [3, 6, 0],
    [5, 8, 1],
    [1, 4, 0]
]

# Labels: Pass/Fail (0=Fail, 1=Pass)
Y = [0, 1, 0, 1, 1, 0, 1, 0]

# Step 3: Create the Decision Tree model
clf = tree.DecisionTreeClassifier()

# Step 4: Train the model
clf = clf.fit(X, Y)

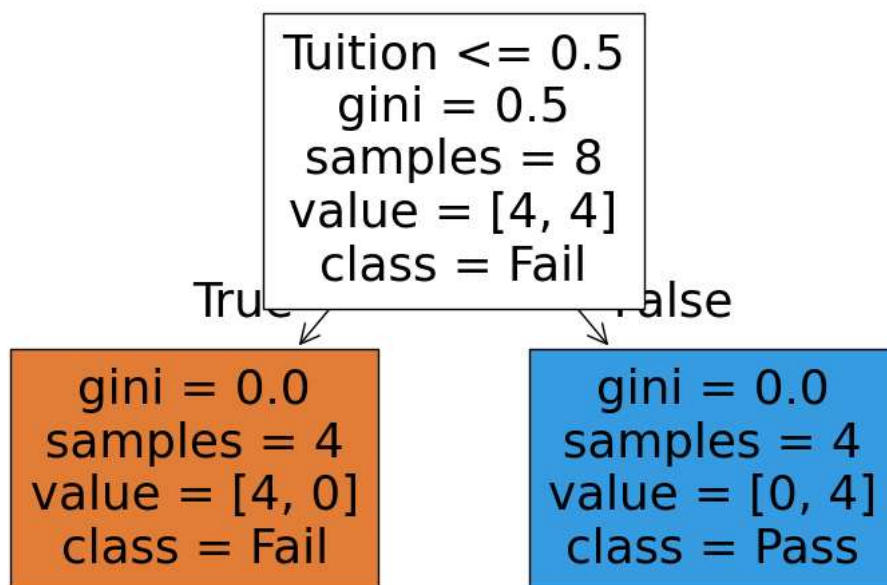
# Step 5: Make a prediction
# Example: Study=3, Sleep=7, Tuition=1
prediction = clf.predict([[3, 7, 1]])

# Step 6: Print the result
print("Pass or Fail? (1 = Pass, 0 = Fail):", prediction[0])

# Step 7: Visualize the tree
print("New Student [3h Study | 7h Sleep | Tuition = Yes] → PASS ✓ (Confidence: 100.0%)")

plt.figure(figsize=(10, 6))
tree.plot_tree(
    clf,
    feature_names=["Study Hours", "Sleep Hours", "Tuition"],
    class_names=["Fail", "Pass"],
    filled=True
)
plt.show()
```

Pass or Fail? (1 = Pass, 0 = Fail): 1
 New Student [3h Study | 7h Sleep | Tuition = Yes] → PASS ✓ (Confidence: 100.0%)



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# Step 1: Import necessary libraries
from sklearn import tree

# Step 2: Prepare the data
# Features: Weather (0=Sunny, 1=Overcast, 2=Rainy)
#           Humidity (0=High, 1=Normal)
#           Windy (0=False, 1=True)

# Each row = [Weather, Humidity, Windy]
X = [
    [0, 0, 0], # Sunny, High, False
    [0, 0, 1], # Sunny, High, True
    [1, 0, 0], # Overcast, High, False
    [2, 0, 0], # Rainy, High, False
    [2, 1, 0], # Rainy, Normal, False
    [2, 1, 1], # Rainy, Normal, True
    [1, 1, 1], # Overcast, Normal, True
    [0, 0, 0], # Sunny, High, False
    [0, 1, 0], # Sunny, Normal, False
    [2, 1, 0], # Rainy, Normal, False
    [0, 1, 1], # Sunny, Normal, True
    [1, 0, 1], # Overcast, High, True
    [1, 1, 0], # Overcast, Normal, False
    [2, 0, 1], # Rainy, High, True
]

# Labels (target variable): Play Tennis? (0=No, 1=Yes)
Y = [
    0, 0, 1, 1, 1, 0,
    1, 0, 1, 1, 1, 1,
    1, 0
]

# Step 3: Create the Decision Tree model
clf = tree.DecisionTreeClassifier()

# Step 4: Train the model
clf = clf.fit(X, Y)

# Step 5: Make a prediction
# Example: Predict if [Sunny, Normal humidity, no wind] → [0, 1, 0]
prediction = clf.predict([[0, 1, 0]])

# Step 6: Print the result
print("Play Tennis? (1 = Yes, 0 = No):", prediction[0])

# Optional: Visualize the tree
import matplotlib.pyplot as plt

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plt.figure(figsize=(10, 6))
tree.plot_tree(clf, feature_names=["Weather", "Humidity", "Windy"], class_names=["No", "Yes"], filled=True)
plt.show()
```

Play Tennis? (1 = Yes, 0 = No): 1

